Computer Organization and Architecture

Assignment 1

1. What is Yotta?

Yotta is an Indian tech company that provides cloud computing solutions and data center services.

Yotta Features:

- Cloud computing services
- Data storage security
- Lower energy consumption in data centers
- Disaster recovery options

Uses:

- Artificial Intelligence (AI)
- Healthcare
- Banking services
- Public services
- Education and research

2. What is NPTEL?

NPTEL (National Programme on Technology Enhanced Learning) is an initiative funded by the Ministry of Education, Government of India, to offer online courses and certifications in different fields.

It provides free online classes, video lectures recorded by IIT professors, and certificates if you pass the final exam.

It helps spread tech awareness, supports us during internships and placements, offers recognized industry certificates, and lets us learn whenever and wherever we want.

Overall, NPTEL is a great platform for students to upgrade their skills and learn straight from top Indian institutions.

3. Who won Turing Award recently? What is the Turing Award?

The A.M. Turing Award is the highest honor in computer science, often referred to as the "Nobel Prize of Computing." It recognizes individuals who have made major contributions to computing and computer science.

The most recent award in 2026 was given to Andrew G. Barto and Richard S. Sutton for their groundbreaking work in Reinforcement Learning, a key area of Artificial Intelligence. Their research helped establish how computers learn through experience—by trying different actions, receiving feedback through rewards or penalties, and improving over time. Their contributions have significantly impacted machine learning and decision-making systems, ranging from robotics to game-playing AI.

4. What is Raspberry-Pi?

Raspberry Pi is a tiny, affordable computer developed by the Raspberry Pi Foundation in the UK. Launched in 2012, it was designed to help students learn programming, computer science, and electronics through hands-on practice. Even though it's roughly the size of a credit card, it can handle basic daily tasks like browsing the web, streaming videos, coding programs, and running an operating system.

It is widely used in projects related to robotics, IoT, home automation, AI, and embedded systems.

5. What are the parameters to decide which programming language should be used to develop an application?

Choosing the right programming language depends on a few key factors:

- Type of application
- Required speed and security
- Platform support
- Ease of development

For example, Python is commonly chosen for AI, Java is widely used for Android applications, and C++ is popular for game development and system software. Picking the right language helps boost performance while saving development time.

6. Why is GPU faster than CPU ?

A Graphics Processing Unit (GPU) handles tasks involving lots of repetitive calculations much faster than a Central Processing Unit (CPU). That's because a GPU contains hundreds or thousands of smaller cores working together at the same time—a process known as parallel processing. This makes it super efficient for graphics rendering, gaming, video editing, and AI tasks. On the other hand, a CPU has fewer but more powerful cores built to execute sequential tasks one by one. So while a CPU handles everyday general computing better, a GPU excels at heavy, parallel workloads.

7. Explain CPU, GPU, TPU.

CPU	GPU	TPU
Stands for Central Processing Unit.	Stands for Graphics Processing Unit.	Stands for Tensor Processing Unit.
Handles general computing tasks.	Focused on graphics and parallel processing tasks.	Specifically designed for Artificial Intelligence workloads.
Has a small number of powerful cores.	Has a large number of smaller processing cores.	Uses specialized AI processing units.
Used for general daily tasks.	Used for graphics rendering and gaming.	Used for machine learning and deep learning applications.

8. RAM Size, Word Size, Speed and Operating System of my Computer:-

RAM Size: 8 GB
Word Size: 64-bit

Speed: 2.10 GHz
Operating System: Windows 64-bit

9. Who is the founder of Intel, AMD?

Intel was founded by Gordon Moore and Robert Noyce.
AMD (Advanced Micro Devices) was founded by Jerry Sanders along with seven co-founders.

10. What is the CPU organization of the processor used in our laptop?

My laptop uses an AMD Ryzen 5 5500U processor. It has a 64-bit x86-64 CPU organization with 6 cores and 12 logical processors. It is a multicore and multithreaded processor, which helps the laptop run many tasks at the same time.

11. What is the Stampeding Herd Problem?

The Stampeding Herd Problem happens when multiple threads or processes are waiting for a single shared resource. The moment that resource opens up, all of them try to access it at the exact same time. This sudden surge overloads the CPU, causing heavy processing overhead and slowing down the system's performance.

You can imagine it like students standing outside a classroom before a lecture. As soon as the teacher opens the door, everyone tries to rush in together. Instead of getting in smoothly, people get stuck in the doorway and everything becomes crowded and slow. The same thing happens in computers when many processes compete for one resource simultaneously.

12. Takeaways from India AI Impact Summit 2026.

- AI should benefit everyone, not just big corporate entities and developed countries.
- Responsible AI must remain fair, safe, transparent, and respectful of user privacy.
- India aims to lead globally in AI by building solutions and tech for the world.
- AI can transform sectors like healthcare, education, farming, finance, and public services.
- Addressing AI challenges requires global cooperation across countries.
- Investing in core infrastructure like compute power and data centers is essential for future growth.
- Upskilling students and working professionals in AI is necessary for upcoming job markets.
- Ethical guidelines and security measures need to be built into AI development right from the start.
- Startups and research centers play a crucial role in driving AI innovation and economic growth.
- AI shouldn't stay confined to research labs; it needs to solve real-world problems and improve everyday life.

13. The Imitation Game

Watching *The Imitation Game* showed me that intelligence alone isn't enough—perseverance and patience matter just as much. The movie highlights how Alan Turing relied on his knowledge and stayed persistent even when people doubted him, proving how one person's ideas can impact the entire world.

It also taught me the value of teamwork. Even with incredible talent, Turing couldn't have achieved his breakthrough without his team. Working together always leads to better results than working alone.